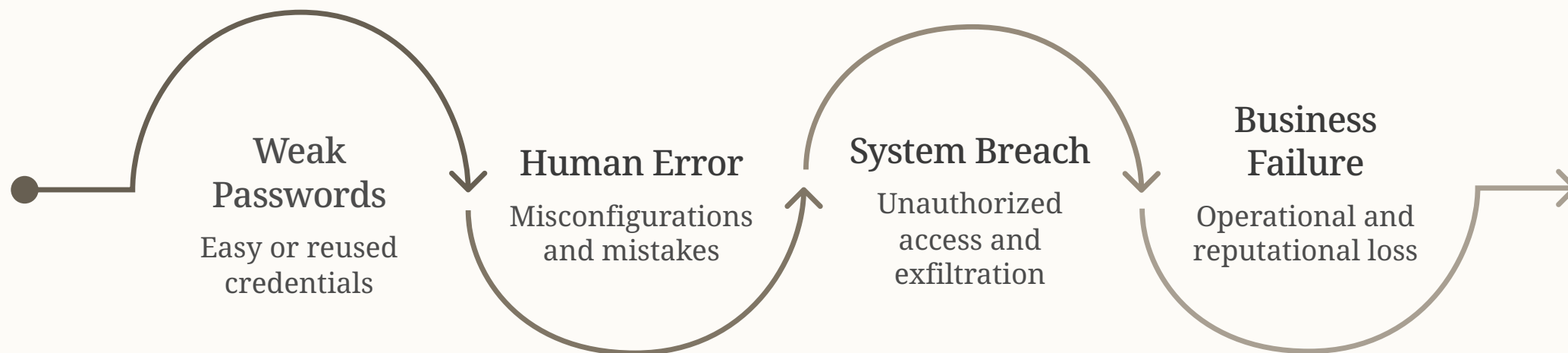


Cryptography & Network Security: Why Modern Systems Keep Failing

Passwords. Ransomware. AI Attacks. Human Mistakes. This presentation cuts through the noise and delivers what security-aware professionals need to know — right now.



A Hospital, Locked Down

What Happened

- MRI machines taken offline
- Staff forced onto paper records
- Ambulances redirected to other hospitals
- Patient care severely compromised

The origin: A single phishing email clicked by one employee.

The Question Every Leader Must Answer

The attackers demanded a ransom to restore systems. Lives hung in the balance. Administrators faced an impossible choice: pay criminals or keep critical care offline.

 Would you pay the ransom? There is no clean answer — only risk tradeoffs with real human consequences.

Cybersecurity Is No Longer "an IT Issue"

Every sector of modern life now runs on digital infrastructure — and every sector is a target. The consequences of a breach extend far beyond a server room.



Banking

Account takeovers and wire fraud cost billions annually. A single compromised credential can drain accounts in minutes.



Healthcare

Ransomware on hospital networks delays surgeries, corrupts patient records, and in documented cases, contributes to patient deaths.



Critical Infrastructure

Airports, power grids, and government systems face constant probing. One misconfiguration away from a national incident.



Your Device

Your phone holds your identity, your finances, and your communications. One weak password can impact millions through a single point of failure.

What Is Computer Security?

Computer security is the practice of protecting systems, data, and identities from unauthorized access, disruption, or destruction — **before damage happens**.

 Security is not a product you buy. It is a posture you maintain — continuously.



Devices

Endpoints, servers, IoT



Data

Files, databases, backups



Networks

Traffic, protocols, access



Identity

Credentials, roles, trust

THE HUMAN PROBLEM

Hackers Rarely Hack Systems First. They Hack People.

The most sophisticated firewall in the world cannot stop an employee who is deceived into handing over their credentials. Social engineering exploits human psychology — not technical vulnerabilities.

→ Urgency

"Your account will be suspended in 24 hours." Panic overrides judgment.

→ Curiosity

"See who viewed your profile." Curiosity triggers clicks without verification.

→ Fear

"Law enforcement has flagged your device." Fear of consequences drives compliance.

→ Authority

"This is your CEO — wire the funds now." Impersonation bypasses skepticism.

LIVE SCENARIO

Your Inbox Just Received This

URGENT: Payroll Error — Action Required

Our records indicate a discrepancy in your direct deposit information. To ensure you receive your paycheck on time, please verify your banking details immediately by clicking the link below. Failure to act within 2 hours may delay your payment.

A. Click the link

Fast. Feels necessary. Probably a mistake.

B. Ignore it entirely

Safe — but what if it's real?

C. Verify the sender first

The only correct answer. Always.

Why C Wins Every Time

Legitimate payroll systems never demand immediate action via email. Call HR directly. Never use contact info embedded in the suspicious message itself.

The Attack Surface Has Never Been Wider

Modern threat actors operate with speed, automation, and scale that outpaces most organizational defenses. These are not hypothetical risks — they are active, daily realities.



Ransomware

Encrypts your data and demands payment. Downtime averages 21 days.



AI Phishing

Personalized, grammatically perfect, generated at massive scale by automated systems.



Deepfakes

Voice and video cloning used to impersonate executives and authorize fraud.



Insider Threats

Disgruntled or compromised employees with legitimate access cause the most costly breaches.



Credential Theft

Stolen usernames and passwords — often purchased on dark web marketplaces for under \$10.



Cloud Compromise

Misconfigured storage buckets and over-permissioned accounts expose millions of records.

CASE STUDY

MGM Resorts: Taken Down by a Phone Call

The Attack Method

Attackers found an MGM employee on LinkedIn, then called the IT help desk impersonating that employee. In under 10 minutes, they obtained credentials sufficient to begin the breach.

No malware. No zero-day exploit. Just a convincing voice and a cooperative help desk.

The Fallout

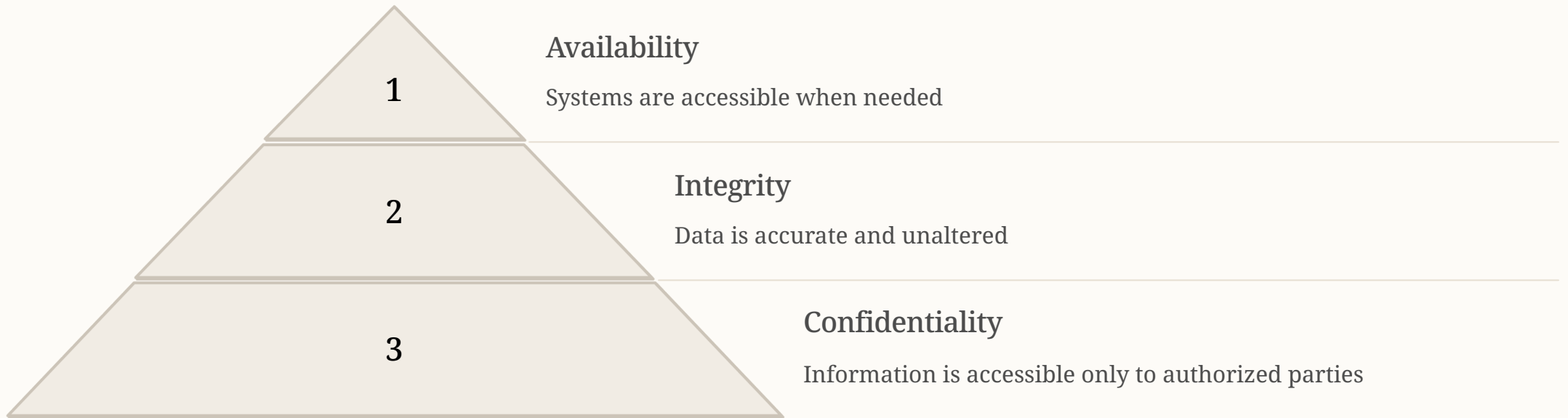
- Casino slot machines went offline across properties
- Hotel room key systems failed
- Estimated losses exceeded \$100 million
- Customer data — including passport numbers — compromised



The most dangerous attack vector is often the one you cannot patch: your own staff.

The CIA Triad: The Foundation of All Security

Every security decision — from password policy to firewall rules — ultimately protects one or more of these three properties. When any one fails, business operations fail with it.



Online Banking: Where All Three Must Hold Simultaneously

Confidentiality

Your account balance, transaction history, and personal details must never be visible to unauthorized users.
Breached by: credential theft, man-in-the-middle attacks, unencrypted connections.

Integrity

A \$500 transfer must arrive as \$500 — not \$5,000. Data in transit must be tamper-proof. Breached by: packet manipulation, SQL injection, fraudulent transaction insertion.

Availability

Banking must be online 24/7. Downtime means lost revenue, regulatory penalties, and customer erosion.
Breached by: DDoS attacks, ransomware, infrastructure failures.

- ❓ Which is hardest to protect? Availability — because attackers only need to overwhelm one point, while defenders must protect every point, all the time.

What Is Cryptography?

The Core Idea

Cryptography is the mathematical science of hiding information. It transforms readable data into an unreadable format — and back again — using algorithms and keys.

Without encryption, every message you send, every password you type, and every transaction you make travels the internet in plain text. Readable by anyone.

Where Cryptography Operates Right Now



Banking & Finance

Every card swipe and wire transfer is cryptographically protected.



Messaging Apps

WhatsApp, Signal — end-to-end encryption means only sender and recipient can read messages.



VPNs & Cloud Storage

Tunnels and vaults built entirely on cryptographic trust.

Security Evolved Because Attackers Evolved

Ancient Rome

Caesar Cipher shifts letters by a fixed number.
Simple substitution — crackable in minutes today.

1

Digital Era

AES, RSA, and TLS protect trillions of transactions daily. Complexity replaced mechanical ingenuity.

3

2

World War II

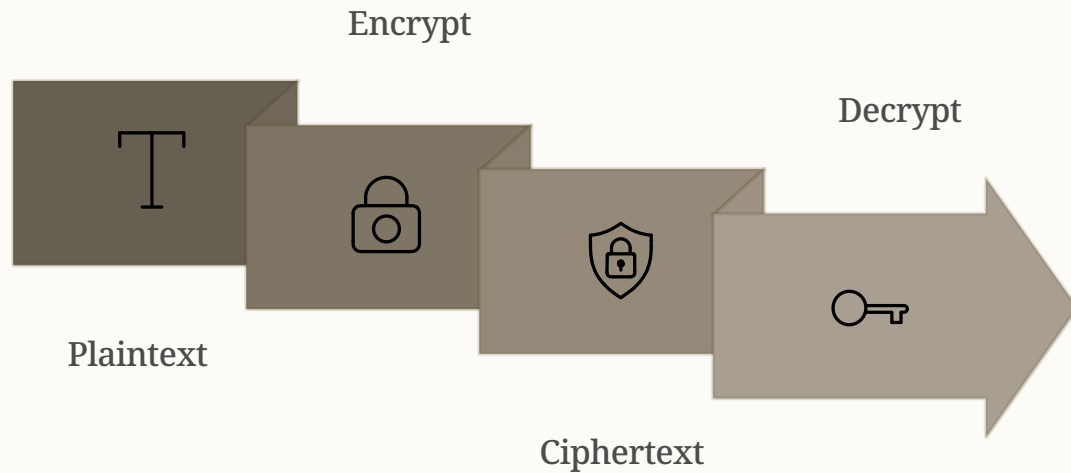
Nazi Germany's Enigma Machine used rotating mechanical keys. Cracking it at Bletchley Park shortened the war by an estimated two years.

4

2026 & Beyond

AI-assisted cryptanalysis and quantum computing threaten to break today's strongest encryption standards.

Encryption, Explained Simply



Encryption converts human-readable data into mathematical noise.
Decryption reverses the process — but only for parties who hold the correct key.

The Critical Principle

The strength of encryption is not in hiding the algorithm — it's in controlling the key. A compromised key renders the strongest cipher completely useless.

What Makes a Key Strong?

- Length: 256-bit vs. 40-bit
- Randomness: Truly unpredictable generation
- Rotation: Changed before it can be cracked
- Storage: Never stored alongside the data it protects

Symmetric Encryption: One Key, Two Jobs

How It Works

A single secret key is used to both encrypt and decrypt data. Both the sender and receiver must possess the same key — and keep it secret from everyone else.

The Standard: AES

The Advanced Encryption Standard (AES-256) is used by the U.S. government, financial institutions, and most enterprise software. At current computing power, brute-forcing AES-256 would take longer than the age of the universe.

Strengths

- Extremely fast — ideal for large data volumes
- Computationally efficient on all hardware
- Well-tested and widely standardized

The Core Problem

If you and I need to share an AES key securely — how do we do it? We can't email it (insecure). We can't mail it (slow). This **key distribution problem** is exactly what asymmetric encryption was invented to solve.

Asymmetric Encryption: The Lock Anyone Can Use, Only You Can Open

1

Public Key

Shared openly with the world. Anyone can use it to encrypt a message to you — like a publicly posted padlock.

2

Encryption

The sender uses your public key to lock the message. Once locked, only the matching private key can open it.

3

Private Key

Held exclusively by the recipient. Never shared. Never transmitted. Its secrecy is the entire basis of trust.

4

Decryption

Only the private key holder can decrypt. This enables HTTPS, digital signatures, and secure email — the backbone of the modern web.

RSA (Rivest–Shamir–Adleman) is the most widely deployed asymmetric algorithm. Every time you see a padlock in your browser, asymmetric encryption is protecting that session.

Coffee Shop Banking: Who Really Failed?

The Scenario

A student connects to an open public WiFi network at a coffee shop and logs into their bank. An attacker on the same network runs a man-in-the-middle attack, intercepting unencrypted session data. No VPN. No MFA. Credentials captured in seconds.

Assigning Responsibility

The User

No VPN, no MFA, no awareness of public network risk. Primary responsibility.

The Coffee Shop

No network isolation between users. A properly segmented network limits interception risk.

The Bank

Should enforce MFA and detect anomalous login locations. Shared accountability.



Answer: All three. Security is a shared responsibility — and the weakest link always determines the outcome.

Cloud ≠ Secure

The Shared Responsibility Model

Cloud providers like AWS, Azure, and Google Cloud secure the underlying infrastructure — physical data centers, network hardware, and hypervisors. **Everything above that layer is your problem.**

 The #1 cause of cloud data breaches is not hacking. It is misconfiguration — S3 buckets left public, admin accounts with no MFA, overly permissive IAM roles.

What You Own in the Cloud

01

Data Classification

Know what you're storing and how sensitive it is.

02

Access Controls

Least-privilege access for every user and service account.

03

Configuration Audits

Continuously scan for exposed storage, open ports, and weak policies.

AI & Cybersecurity: The Double-Edged Weapon

Artificial intelligence is simultaneously the most powerful defensive tool available — and the most dangerous offensive capability ever handed to attackers.

AI Defending

- Detects anomalous behavior in real time
- Analyzes millions of log events per second
- Predicts attack patterns before execution
- Automates incident response at machine speed

AI Attacking

- Generates hyper-personalized phishing at scale
- Clones voices for CEO fraud and wire transfer scams
- Creates polymorphic malware that rewrites itself to evade detection
- Automates vulnerability scanning across millions of targets simultaneously

❌ The next frontier: AI vs. AI cyber warfare — where autonomous attack agents probe and exploit systems faster than any human team can respond.

Why Organizations Keep Getting Breached

The technology to prevent most breaches exists today. The failures are almost never technical — they are operational, cultural, and procedural.



Human Error

Of breaches involve a human element — phishing, misuse, or mistake.



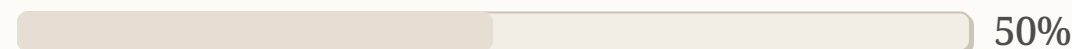
No MFA

Of compromised accounts had no multi-factor authentication enabled at time of breach.



Misconfiguration

Of cloud breaches stem directly from misconfigured access settings — not external exploits.



Unpatched Systems

Of exploited vulnerabilities had patches available but not yet applied at time of attack.

Cybersecurity Is Not About Perfection. It's About Resilience.



Reduce Risk

Eliminate the low-hanging fruit: enforce MFA, patch regularly, and train people to recognize social engineering.



Detect Faster

The average breach goes undetected for 194 days. Speed of detection is the single biggest factor in limiting damage.



Recover Quickly

Tested backup and incident response plans are the difference between a bad week and a company-ending event.

The Final Question

What is more dangerous to your organization: a sophisticated hacker — or the employee who uses **"Password123"** on every account and clicks every urgent email?

The answer shapes your entire security strategy. Invest in people first. Technology second. Process always.